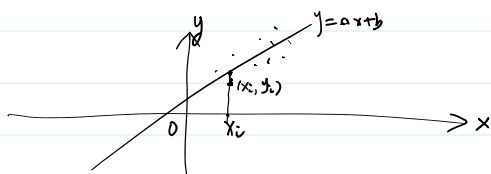


第10讲. 条件极值.

作业: P95. 9. 10 (5), (6). 12 (1).

练习1. 设 $f(x, y) \in C^2(\mathbb{R}^2)$. $\forall \theta$, 记 $g_\theta(t) = f(t \cos \theta, t \sin \theta)$ 若对 $\forall \theta$, 有 $\left. \frac{dg_\theta(t)}{dt} \right|_{t=0} = 0$ 且 $\left. \frac{d^2 g_\theta(t)}{dt^2} \right|_{t=0} > 0$.证明: $f(0,0)$ 是 $f(x,y)$ 的极小值.分析: ① $f'_x(0,0) = 0$, $f'_y(0,0) = 0$.② $H_f(0,0)$ 正定.例1. 设实验观测数据 $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. ($x_i \neq x_j$, $i \neq j$)问: 如何由这组数据得到最佳线性近似公式 $y = ax + b$?

$$f(a, b) = \sum_{i=1}^n |ax_i + b - y_i|^2 \quad \text{最小.} \quad \mathbb{R}^2.$$

$$0 = f'_a(a, b) = 2 \sum_{i=1}^n (ax_i + b - y_i) \cdot x_i$$

$$0 = f'_b(a, b) = 2 \sum_{i=1}^n (ax_i + b - y_i)$$

$$\begin{cases} \left(\sum_{i=1}^n x_i^2 \right) a + \left(\sum_{i=1}^n x_i \right) b = \sum_{i=1}^n x_i y_i \\ \left(\sum_{i=1}^n x_i \right) a + n b = \sum_{i=1}^n y_i \end{cases}$$

$$\begin{pmatrix} \sum_{i=1}^n x_i^2 & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & n \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n x_i y_i \\ \sum_{i=1}^n y_i \end{pmatrix}$$

$$n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 > 0.$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n x_i^2 & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^n x_i y_i \\ \sum_{i=1}^n y_i \end{pmatrix}$$

$$A = f''_{aa}(a, b) = 2 \left(\sum_{i=1}^n x_i^2 \right)$$

$$B = f''_{ab}(a, b) = 2 \left(\sum_{i=1}^n x_i \right)$$

$$C = f''_{bb}(a, b) = 2n$$

$$AC - B^2 > 0.$$

$$A > 0$$

$\therefore (a, b)$ 为 $f(a, b)$ 极小值点.

(a, b) 为 $f(a, b)$ 最小值点, 最小 = 验证.

= 条件极值.

求 $f(x, y)$ 在 $\varphi(x, y) = 0$ 下极值

↓
目标函数

↓
约束条件

$$f(x, y), \varphi(x, y) \in C^1, \quad \varphi'_y(x, y) \neq 0.$$

$$\varphi(x, y) = 0 \text{ 确定 } y = y(x) \Rightarrow y'_x = - \frac{\varphi'_x}{\varphi'_y}$$

求 $g(x) = f(x, y(x))$ 极值.

$$g'(x) = 0.$$

||

$$f'_x + f'_y y'_x \Rightarrow y'_x = - \frac{f'_x}{f'_y}$$

$$\Rightarrow \frac{f'_x}{f'_y} = \frac{\varphi'_x}{\varphi'_y}$$

||

$$\frac{f'_x}{\varphi'_x} = \frac{f'_y}{\varphi'_y} = \lambda$$

1. 拉格朗日乘子法

$$1^\circ. \quad \mathcal{L}(x, y) = f(x, y) - \lambda \varphi(x, y) \quad \text{拉格朗日函数.}$$

↓
拉格朗日乘子.

$$2^\circ \text{ 求解 } \begin{cases} \mathcal{L}'_x = f'_x - \lambda \varphi'_x = 0 \\ \mathcal{L}'_y = f'_y - \lambda \varphi'_y = 0 \\ \varphi(x, y) = 0 \end{cases}$$

3°. 求驻点 $P_0(a, b)$.

4°. 判断 $f(x, y)$ 在 (a, b) 条件极值.

$$\forall (x, y), \quad \varphi(x, y) = 0.$$

$$(x, y) \rightarrow (a, b)$$

$$f(x, y) - f(a, b) = \mathcal{L}(x, y) - \mathcal{L}(a, b)$$

$$= \frac{1}{2} \left[\mathcal{L}''_{xx}(P_0)(x-a)^2 + 2\mathcal{L}''_{xy}(P_0)(x-a)(y-b) + \mathcal{L}''_{yy}(P_0)(y-b)^2 \right] + o((x-a)^2 + (y-b)^2)$$

$$\underbrace{(x-a, y-b)}_{\parallel} H_L(a, b) \begin{pmatrix} x-a \\ y-b \end{pmatrix}$$

① $H_2(a, b)$ 正定.

$$f(x, y) > f(a, b).$$

$\therefore (a, b)$ 为 $f(x, y)$ 条件极小值点.

② $H_2(a, b)$ 负定. $f(x, y) < f(a, b)$

$\therefore (a, b)$ 为 $f(x, y)$ 条件极大值点

③ 若 $H_2(a, b)$ 半定. 不定.

$$d^2 L(a, b) = L''_{xx}(a, b) dx^2 + 2L''_{xy}(a, b) dx dy + L''_{yy}(a, b) dy^2.$$

dx 与 dy 关系由对 $\varphi(x, y) = 0$ 微分得到. $\varphi'_x dx + \varphi'_y dy = 0.$

③a) 当 $d^2 L(a, b) > 0$. 则 (a, b) 为 $f(x, y)$ 条件极小值点.

③b) 当 $d^2 L(a, b) < 0$. 则 ... 大 ...

几何解释: 求 $f(x, y)$ 在 $\varphi(x, y) = 0$ 极值.

$$\begin{aligned} &\Downarrow \\ &\underline{y = y(x)}. \quad \text{曲线 } l \text{ 光滑.} \quad \begin{cases} x = x(t) \\ y = y(t) \end{cases} \quad t \in I. \end{aligned}$$

$\vec{c} = (x'(t), y'(t))$ 为 l 切向量.

设 $g(t) = f(x(t), y(t))$ 在 t_0 取得极值.

$$\begin{aligned} x_0 &= x(t_0) \\ y_0 &= y(t_0) \end{aligned} \quad p_0(x_0, y_0) \in l.$$

$$\therefore g'(t_0) = 0.$$

$$\Downarrow \\ f'_x(p_0) x'(t_0) + f'_y(p_0) y'(t_0) = 0$$

$$\underline{\text{grad} f(p_0)} \cdot \vec{c} = 0$$

$$\underline{\text{grad} f(p_0)} \parallel \text{grad} \varphi(p_0) \quad \therefore \text{grad} f(p_0) = \lambda \text{grad} \varphi(p_0)$$

定理 2. 设 $f(x_1, x_2, \dots, x_n), \varphi_i(x_1, \dots, x_n) \ (i=1, 2, \dots, m, m < n) \in C^1$.

求目标函数 $f(x_1, x_2, \dots, x_n)$ 在约束条件 $\begin{cases} \varphi_1(x_1, \dots, x_n) = 0 \\ \vdots \\ \varphi_m(x_1, \dots, x_n) = 0 \end{cases}$ 极值.

$$\text{令 } L(x_1, x_2, \dots, x_n) = f(x_1, \dots, x_n) - \lambda_1 \varphi_1(x_1, \dots, x_n) - \lambda_2 \varphi_2(x_1, \dots, x_n) - \dots - \lambda_m \varphi_m(x_1, \dots, x_n)$$

设 $P_0(x_0, x_0, \dots, x_0)$ 满足约束

$$\begin{cases} \mathcal{L}'_{x_1} = f'_{x_1} - \lambda_1 \varphi'_{1x_1} - \dots - \lambda_m \varphi'_{mx_1} = 0 \\ \vdots \\ \mathcal{L}'_{x_n} = f'_{x_n} - \lambda_1 \varphi'_{1x_n} - \dots - \lambda_m \varphi'_{mx_n} = 0 \\ \varphi_1(x_1, \dots, x_n) = 0 \\ \vdots \\ \varphi_m(x_1, \dots, x_n) = 0 \end{cases}$$

例①若 $H_2(p_0)$ 正定, 则 f 在 p_0 条件极小值.

②若 $H_2(p_0)$ 负定 则 f 在 p_0 条件极大值.

③若 $H_2(p_0)$ 半定或不定.

$$d^2 \mathcal{L}(p_0) = (dx_1, dx_2, \dots, dx_n) H_2(p_0) \begin{pmatrix} dx_1 \\ dx_2 \\ \vdots \\ dx_n \end{pmatrix}$$

$dx_1, dx_2, \dots, dx_n$ 满足关系, 由对 $\varphi_i(x_1, \dots, x_n) = 0$ 微分得到

$$\varphi'_{ix_1} dx_1 + \varphi'_{ix_2} dx_2 + \dots + \varphi'_{ix_n} dx_n = 0, \quad i=1, 2, \dots, m.$$

当 $d^2 \mathcal{L}(p_0) > 0$ 时, 则 f 在 p_0 条件极小值.

当 $d^2 \mathcal{L}(p_0) < 0$ 时, 则 f 在 p_0 条件极大值.

例1. 求 $f(x, y) = xy$ 在 $D = \{(x, y) \mid \frac{x^2}{8} + \frac{y^2}{2} \leq 1\}$ 上最值.

解: 1° $\begin{cases} 0 = f'_x = y \\ 0 = f'_y = x \end{cases} \quad (0, 0).$

$$A = f''_{xx}(0, 0) = 0.$$

$$B = f''_{xy}(0, 0) = 1.$$

$$C = f''_{yy}(0, 0) = 0.$$

$$AC - B^2 = -1 < 0.$$

$(0, 0)$ 不是 $f(x, y)$ 极值点

2° 由 $f(x, y) = xy$ 在 $\frac{x^2}{8} + \frac{y^2}{2} = 1$ 上求极值.

$$\mathcal{L}(x, y) = xy - \lambda \left(\frac{x^2}{8} + \frac{y^2}{2} - 1 \right)$$

$$\mathcal{L}'_x = y - \frac{1}{4} \lambda x = 0$$

$$\begin{cases} \mathcal{L}'_x = y - \frac{1}{4}\lambda x = 0 \\ \mathcal{L}'_y = x - \frac{1}{4}\lambda y = 0 \\ \frac{x^2}{8} + \frac{y^2}{2} = 1 \end{cases} \Rightarrow \frac{y}{x} = \frac{1}{4}\lambda = \frac{1}{x} \Rightarrow \lambda = \pm 2.$$

$$\begin{aligned} xy &= \frac{1}{4}\lambda x^2 \\ xy &= \frac{1}{4}\lambda y^2 \end{aligned} \Rightarrow 2xy = (\frac{1}{4}x^2 + y^2)\lambda = 2\lambda$$

$$\therefore xy = \lambda$$

$$\therefore f_{\max} = 2, \quad f_{\min} = -2.$$

例2. 设 $x > 0, y > 0, z > 0$. 证明: $\sqrt[3]{xyz} \leq \frac{x+y+z}{3}$. 且等号成立 $\Leftrightarrow x=y=z$.

求 $f(x, y, z) = xyz$ 在 $x+y+z=a$. ($\forall a > 0$) 极值.

解: $\mathcal{L}(x, y, z) = \ln x + \ln y + \ln z - \lambda(x+y+z-a)$

$$\begin{cases} \mathcal{L}'_x = \frac{1}{x} - \lambda = 0 \\ \mathcal{L}'_y = \frac{1}{y} - \lambda = 0 \\ \mathcal{L}'_z = \frac{1}{z} - \lambda = 0 \\ x+y+z=a \end{cases} \Rightarrow \begin{aligned} &x=y=z=\frac{1}{\lambda} \\ &\Rightarrow (x, y, z) = \left(\frac{a}{3}, \frac{a}{3}, \frac{a}{3}\right) \\ &\downarrow \\ &p_0. \end{aligned}$$

$$\mathcal{L}''_{xx} = -\frac{1}{x^2}, \quad \mathcal{L}''_{xy} = 0, \quad \mathcal{L}''_{xz} = 0.$$

$$\mathcal{L}''_{yy} = -\frac{1}{y^2}, \quad \mathcal{L}''_{yz} = 0$$

$$\mathcal{L}''_{zz} = -\frac{1}{z^2}$$

$$\therefore H_2(p_0) = \begin{pmatrix} -\frac{1}{x^2} & & \\ & -\frac{1}{y^2} & \\ & & -\frac{1}{z^2} \end{pmatrix}_{p_0} \quad \text{负定.}$$

$$\therefore \underset{x, y, z}{f(x, y, z)} \leq f\left(\frac{a}{3}, \frac{a}{3}, \frac{a}{3}\right) = \left(\frac{a}{3}\right)^3 = \left(\frac{x+y+z}{3}\right)^3$$

$$\frac{1}{xy^2} = \frac{1}{x} \cdot \frac{1}{y^2} = \frac{1}{x} \cdot \left(\frac{1}{y} \right)^2$$

$$\therefore \sqrt[3]{xy^2} \leq \frac{x+y+z}{3}$$

例3. 设 $z = z(x, y)$, 由 $(x^2+y^2)z + \ln z + 2(x+y+1) = 0$ 确定. 求 $z = z(x, y)$ 极值.

$$\begin{aligned} z &= f(x, y) & P_0(x_0, y_0) & & dz &= 0 \\ d^2f(P_0) &= f''_{xx}(P_0) dx^2 + 2f''_{xy}(P_0) dx dy + f''_{yy}(P_0) dy^2 \\ &= (dx, dy) H_f(P_0) \begin{pmatrix} dx \\ dy \end{pmatrix} \end{aligned}$$

$$(2x dx + 2y dy) z + (x^2+y^2) dz + \frac{1}{z} dz + 2(dx+dy) = 0$$

$$(2xz+2) dx + (2yz+2) dy + (x^2+y^2+\frac{1}{z}) dz = 0$$

令 $dz=0$.

$$\begin{cases} 2xz+2=0 \\ 2yz+2=0 \end{cases} \Rightarrow \underline{xz=yx=-1.}$$

$$x=y=-\frac{1}{z}$$

$$\underline{\ln z - \frac{2}{z} + 2 = 0.}$$

$$z=1$$

$$x=y=-1$$

$$z(1,1)=1.$$

$$\begin{aligned} (2z dx + 2x dz) dx + (2z dy + 2y dz) dy + (2x dx + 2y dy - \frac{1}{z^2} dz) dz \\ + (x^2+y^2+\frac{1}{z}) dz^2 = 0 \end{aligned}$$

$$dz=0$$

$$2dx^2 + 2dy^2 + 3dz^2(1,1) = 0$$

$$\therefore d^2z(1,1) = -\frac{2}{3}(dx^2+dy^2) < 0.$$

$\therefore (1,1)$ 为 $z = z(x, y)$ 极大值点. 且极大值为 $z(1,1)=1$.

三. 含参积分.

定积分 $\int_a^b f(x) dx$ $\int f(x)$ 有界 $\rightarrow$ 无界. $\int_0^1 \frac{1}{\sqrt{x}} dx$ — 瑕积分

$$\text{定积分 } \int_a^b f(x) dx \begin{cases} f(x) \text{ 有界} \rightarrow \text{无界} & \int_0^1 \frac{1}{\sqrt{x}} dx \text{ --- 瑕积分} \\ [a, b] \rightarrow [a, +\infty) \text{ 或 } (-\infty, b], (-\infty, +\infty) & \text{广义积分 (反常)} \end{cases}$$

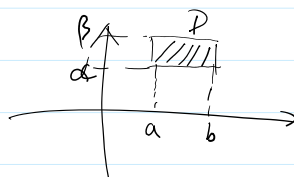
$$\forall u \in [\alpha, \beta], \quad \int_a^{+\infty} f(x) dx \text{ --- 无穷积分}$$

$$\text{定积分: } \int_a^b f(x) dx \rightarrow \int_a^b \underline{f(x, u)} dx = \underline{\varphi(u)} \text{ --- 含参定积分}$$

$$\text{广义积分: } \int_a^{+\infty} f(x) dx \rightarrow \int_a^{+\infty} \underline{f(x, u)} dx \text{ 收敛}$$

$$\parallel \parallel \varphi(u)$$

$$\lim_{P \rightarrow +\infty} \int_a^P f(x) dx \text{ 存在.}$$



1. 含参定积分.

定义1. 设 $f(x, u)$ 定义 $D = [a, b] \times [\alpha, \beta]$ 上. 对 $\forall u \in [\alpha, \beta]$.

若 $f(x, u) \in R[a, b]$. 则 $\varphi(u) = \int_a^b f(x, u) dx$ 为含参变量 u 定积分.

定义1. 设 $f(x, u) \in C(D)$. $D = [a, b] \times [\alpha, \beta]$.

$$\text{则 } \varphi(u) = \int_a^b f(x, u) dx \in C[\alpha, \beta].$$

$$\text{分析: } \forall u_0 \in [\alpha, \beta]. \quad \lim_{u \rightarrow u_0} \varphi(u) = \varphi(u_0)$$

$$\because f(x, u) \in C(D).$$

$$\therefore \forall \varepsilon > 0. \exists \delta > 0. \text{ s.t. } \forall (x, u) \in D. (x, u_0) \in D. \text{ 当 } |u - u_0| < \delta \text{ 时 } |f(x, u) - f(x, u_0)| < \frac{\varepsilon}{b-a}$$

$$|\varphi(u) - \varphi(u_0)| \leq \int_a^b |f(x, u) - f(x, u_0)| dx < \varepsilon.$$

$$\therefore \varphi(u) \in C[\alpha, \beta].$$

$$\lim_{u \rightarrow u_0} \varphi(u) = \varphi(u_0) = \int_a^b f(x, u_0) dx$$

$$\lim_{u \rightarrow u_0} \int_a^b \underline{f(x, u)} dx = \int_a^b (\lim_{u \rightarrow u_0} \underline{f(x, u)}) dx$$

$$\text{例1. 求 } \lim_{y \rightarrow 0^+} \int_0^1 \frac{1}{1 + (1+x)y} dx = \int_0^1 \frac{1}{1+e^x} dx = \int_0^1 \frac{e^{-x}}{1+e^x} dx = - \int_0^1 \frac{d(e^{-x}+1)}{e^{-x}+1}$$

$$f(x, u) = \frac{1}{1 + (1+x)y}$$

$$u \neq 0$$

$$= -\ln(e^{-x}+1) \Big|_0^1$$

$$f(x, y) = \begin{cases} \frac{1}{1 + (1+xy)^{\frac{1}{y}}} & y \neq 0 \\ \frac{1}{1+e^x} & y = 0 \end{cases}$$

$$\lim_{y \rightarrow 0^+} f(x, y) = \frac{1}{1+e^x}$$

$$y \neq 0$$

$$y = 0$$

$$= -\ln(e^{-x} + 1) \Big|_0^1$$

$$= \ln 2 - \ln(1+e^{-1}) = \ln \frac{2e}{e+1}$$

$$f(x, y) \in C(D)$$

$$D = [0, 1] \times [0, 1]$$

定理 2. 设 $f(x, u)$ 定义在 $D = [a, b] \times [\alpha, \beta]$. 满足

$$① \quad \forall u \in [\alpha, \beta], \quad f(x, u) \in R[a, b].$$

$$② \quad f'_u(x, u) \in C'(D).$$

则 $\varphi(u) = \int_a^b f(x, u) dx$ 在 $[\alpha, \beta]$ 上可导 且 $\varphi'(u) = \int_a^b f'_u(x, u) dx$.

$$\frac{d \left(\int_a^b f(x, u) dx \right)}{du}$$

证: $\forall u_0 \in [\alpha, \beta]$.

$$\lim_{\Delta u \rightarrow 0} \frac{\varphi(u_0 + \Delta u) - \varphi(u_0)}{\Delta u} = \lim_{\Delta u \rightarrow 0} \frac{\int_a^b (f(x, u_0 + \Delta u) - f(x, u_0)) dx}{\Delta u}$$

$$\parallel$$

$$\varphi'(u_0)$$

$$\stackrel{\exists \theta \in (0, 1)}{=} \lim_{\Delta u \rightarrow 0} \int_a^b \underbrace{f'_u(x, u_0 + \theta \Delta u)}_{\parallel} dx$$

$$= \int_a^b f'_u(x, u_0) dx$$